

Day 15 - Auto Scaling Group + ALB + RDS - SAA-C03 Deep Dive - Solomon Igein (573705805662)

Account: 573705805662 | Region: us-east-1 | VPC: vpc-0a9725e6138ed7268 | Date: Aug 11, 2026

1. Architecture You Built (CREATE_COMPLETE 39 resources)

Stack: day15-stack-v2 -> CREATE_COMPLETE

User -> ALB (day15-alb-cfn, internet-facing, 2 AZs subnet-0e60ad810626d2cfa az4 + subnet-0b78aa88fec02c16c az3) -> Target Group day15-tg-cfn (Port 80, HealthCheck /) -> ASG day15-asg-cfn (Min 1 Max 5 Desired 2, HealthCheckType ELB, Grace 120s) -> Launch Template day15-lt-golden-cfn (ami-0c02fb55956c7d316, t3.micro, mykey3.pem) -> EC2 httpd + mysql client -> RDS day15-db-cfn.ckh2o0wk0dwn.us-east-1.rds.amazonaws.com (db.t3.micro, mysql 8.0, 20GB gp2, private, SG 3306 only from Instance SG)

3-Tier SG Model: ALBSG: 80 0.0.0.0/0 -> InstanceSG: 80 from ALBSG + 22 0.0.0.0/0 -> RDSSG: 3306 from InstanceSG. Principle of least privilege.

UserData Proof: Visits count in RDS: 6 -> 164 PERSISTS after terminate! Stateless EC2 vs Stateful RDS.

2. Core Concepts for SAA-C03

ASG: Desired vs Min/Max, ELB health check vs EC2, Termination policies (OldestLaunchTemplate), Launch Template vs Config, Scheduled Actions (scale-out 8am Desired 2, scale-in 8pm Desired 1) for cost optimization.

ALB: Application Load Balancer L7, Listener 80 HTTP -> Forward TG, TG health checks deregister unhealthy, Cross-AZ enabled, Internet-facing needs public subnets with IGW.

RDS: db.t3.micro Free Tier, DBSubnetGroup needs 2 AZs, MultiAZ false for dev, BackupRetention 0 for lab, PubliclyAccessible false, DeletionProtection false for CloudFormation cleanup. EngineVersion must be generic 8.0 not 8.0.36 (region-specific availability).

CloudFormation: ROLLBACK_COMPLETE = failed, check Events -> Status Reason bottom-up, DELETE before recreate, Outputs tab contains ALBDNS not Stacks list page.

3. Troubleshooting You Did (Real DevOps)

Error Screenshot	Root Cause	Fix
Cannot find version 8.0.36 for MySQL	Specific minor version not available in us-east-1	EngineVersion: '8.0' (auto-resolve to 8.0.40)
KeyName day2-key does not exist	External Lab has mykey3.pem, saa-keypairs does not	Use mykey3.pem from EC2 > Key pairs
ALBDNS not found	Looked at Stacks (0) list	Click stack name -> Outputs tab -> ALBDNS
Yahoo search page	Pasted ALB DNS into search engine	Paste http://ALBDNS into browser address bar
Visits count resets?	If data on EBS it would reset	RDS persists: 6 -> 164 after 2 terminations

4. Chaos Test - Why Visits 164 Proves Exam Concept

You terminated i-05deabd50aa7d51c1 (Running) -> Successfully terminated. ASG Activity: Terminating + Launching. New instance i-0500a471600465154 3.87.102.85 Running 3/3 checks. ALB auto-registered. Refresh http://ALB -> New IP ip-172-31-19-210 (172.31.19.210) but Visits 164 (not 1). This proves Decoupling: EC2 stateless, RDS stateful, ALB provides HA, ASG provides FT & elasticity. Interview answer: 'State in managed DB, not instance store.'

5. Cost - Why Bill USD 0.00

t3.micro 750hrs/mo Free Tier, db.t3.micro 750hrs/mo Free Tier, 20GB gp2 Free Tier, ALB ~0.0225\$/hr + LCU. Learner Lab 573705805662 has credits, but you kept 0 by deleting stack immediately after chaos. STACKS 0, INSTANCES Terminated, DATABASES 0. Always CloudFormation Delete, not manual EC2 terminate (ASG would relaunch).

6. SAA-C03 Exam Questions - Day 15

Q1: EC2 in ASG keeps terminating, what to check first?

A: ASG Activity + TG Health Checks. If UserData fails (mysql install fails), TG marks unhealthy -> ASG terminates. Check ELB HealthCheckGracePeriod 120s. Your case: 164 visits proves UserData succeeded.

Q2: ALB returns 502 Bad Gateway after deploy, cause?

A: InstanceSG not allowing 80 from ALBSG, or TG health check path / returning 404. Your SG: InstanceSG 80 from ALBSG correct, UserData creates /var/www/html/index.html -> 200.

Q3: RDS Cannot find version 8.0.36 - fix?

A: AWS region doesn't have that exact minor. Use major version 8.0 or omit. You fixed: EngineVersion '8.0' -> resolved to 8.0.40. Also check DBSubnetGroup needs 2 AZs.

Q4: Need to keep data when ASG scales in, where store?

A: NOT on EC2 EBS/instance store. Use RDS (your solution), EFS, S3, DynamoDB. Your proof: Visits count in RDS: 164 PERSISTS after terminate.

Q5: How to make ASG cost optimized for 8am-8pm workload?

A: Scheduled Actions: scale-out-8am Recurrence 0 8 * * * Desired 2, scale-in-8pm 0 20 * * * Desired 1. Plus Target Tracking on CPU 50%. You implemented both scheduled actions.

Q6: CloudFormation ROLLBACK_COMPLETE - what to do?

A: Check Events tab bottom-up for first CREATE_FAILED Status Reason, fix template, Delete stack (cannot update rollback), recreate. You did: Found DB version error, Deleted day15-stack15d5, recreated v2 -> CREATE_COMPLETE.

Q7: ALB DNS pasted but shows Yahoo search, fix?

A: User pasted into search engine not address bar. Must use http://ALBDNS in address bar, port 80. ALB is HTTP, not HTTPS unless ACM cert.

Q8: Why 3 Security Groups?

A: Defense in depth. ALBSG public 80, InstanceSG only from ALBSG (not 0.0.0.0/0 for 80), RDSSG only 3306 from InstanceSG. If InstanceSG allowed 3306 0.0.0.0/0 would fail audit.

Q9: After CFN delete, EC2 still running?

A: If you manually terminated EC2 not via CFN, ASG launches new (desired capacity). Must Delete stack via CloudFormation, which deletes ASG first -> no launch. Your cleanup: Stacks 0, Instances all Terminated, DB 0 -> perfect.

Q10: Visits count 6 -> 164, what does it mean?

A: 158 additional inserts, each refresh does INSERT INTO visits. Survived 2 terminations (i-02a85f... shutting-down, i-05deabd... terminated) -> new IP 172.31.19.210. Proves RDS MultiAZ false still persists, AZ failure would lose if not MultiAZ, but instance failure safe.